

Morning Paper Report - Friday, September 11, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Similar Quality of Visual Working Memory Representations between Negative and Positive Attentional Templates.

Chidharom M, Zafarmand M, Carlisle NB
Journal of cognition, 2024

Source: PubMed; matched terms: visual working memory, working memory representation, working memory

Link: <https://pubmed.ncbi.nlm.nih.gov/39035075/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

2. Linking behavioral and neural measures of visual search

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual search, computational model, computational models, modeling, attentional guidance

Link: https://osf.io/preprints/psyarxiv/rxy7f_v1/

Goal: Visual search is a ubiquitous behavior, yet its neural mechanisms remain incompletely understood

Method: Here, we combine behavioral modeling and fMRI to identify neural signatures of core visual search computations

Results: We show that behavioral search slopes – quantifying how response time scales with set size – are mirrored by corresponding fMRI-derived slopes, such that neural responses increase systematically with set size and scale with target-distractor similarity

Conclusion: Finally, we show that fMRI responses in bidimensional search can be predicted from activity measured during unidimensional search, demonstrating that computational models of behavior generalize to neural data Together, these findings establish a quantitative link between behavioral and neural measures of visual search and provide a framework for studying attentional guidance and feature integration in the human...

3. More than Just Measurement Models? Processing Insights from Population Coding Models of Visual Working Memory Retrieval

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory

Link: https://osf.io/axcmh_v2/

Goal: Recent studies of visual working memory (VWM) have investigated a population coding framework, in which items in memory are coded by an array of noisy, tuned detectors

Method: Studies have compared alternative tuning functions and detector noise distributions, finding differences in fit in some cases and substantial model mimicry in others

Results: Within this framework, heavy-tailed distributions of memory errors found in continuous-outcome VWM tasks reflect noise in the detector array rather than guessing

Conclusion: We compared four population coding models and found that: (1) a von Mises tuning function provided better model fit and more stable estimates than a similarity-based, exponential-decay tuning function; (2) normal and Gumbel noise functions provided comparably good fits, but different patterns of parameter estimates; (3) Depending on task complexity, the amplitude of tuning function followed either an inverse...

4. Biologically Inspired Computational Models of Visuospatial Working Memory: A Systematic Review

Viviana Dueñas, Sonia López, José-Antonio Cervantes, Gerardo Ortiz Torres
2025

Source: Crossref; matched terms: working memory, computational model, computational models

Link: <https://doi.org/10.2139/ssrn.5357648>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

5. A General Closed-loop Predictive Coding Framework for Auditory Working Memory

Zhongju Yuan, Geraint Wiggins, Dick Botteldooren
2025 International Joint Conference on Neural Networks (IJCNN), 2025

Source: Crossref; matched terms: working memory, neural network, neural networks, predictive coding

Link: <https://doi.org/10.1109/ijcnn64981.2025.11229059>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.